NAME: GALLA MOHAN SAI

MODULE NO : 2

Exercise 2: E-commerce Platform Search Function

Product.java

```
class Product {

   int productId;
   String productName;
   double price;

   Product(int productId, String productName, double price) {
      this.productId = productId;
      this.productName = productName;
      this.price = price;
   }

   public void display() {
      System.out.println("Product ID: " + productId
            + ", Product Name: " + productName
            + ", Price: ₹" + price);
   }
}
```

LinearSearch.java

```
class LinearSearch {

   public static Product search(Product[] products,
                        int targetId) {

      for(Product product : products) {

         if(product.productId == targetId) {
            return product;
         }
      }

      return null;
```

```
    }
}
```

BinarySearch.java

```
class BinarySearch {

    public static Product search(Product[] products,
                                 int targetId) {

        int low = 0;
        int high = products.length - 1;

        while(low <= high) {

            int mid = (low + high) / 2;

            if(products[mid].productId == targetId) {
                return products[mid];
            }

            if(products[mid].productId < targetId) {
                low = mid + 1;
            }
            else {
                high = mid - 1;
            }
        }

        return null;
    }
}
```

EcommerceSearch.java

```
public class EcommerceSearch {

    public static void main(String[] args) {

        Product[] products = {

            new Product(201, "Keyboard", 799),
            new Product(202, "Mouse", 499),
            new Product(203, "Monitor", 8999),
```

```
            new Product(204, "Headphones", 1499),
            new Product(205, "Webcam", 1999)

        };

        int targetId = 204;

        System.out.println("LINEAR SEARCH RESULT");

        Product result1 =
                LinearSearch.search(products, targetId);

        if(result1 != null) {
            result1.display();
        }
        else {
            System.out.println("Product Not Found");
        }

        System.out.println();

        System.out.println("BINARY SEARCH RESULT");

        Product result2 =
                BinarySearch.search(products, targetId);

        if(result2 != null) {
            result2.display();
        }
        else {
            System.out.println("Product Not Found");
        }
    }
}
```

OUTPUT:

```
LINEAR SEARCH RESULT
Product ID: 204, Product Name: Headphones, Price: ₹1499.0

BINARY SEARCH RESULT
Product ID: 204, Product Name: Headphones, Price: ₹1499.0
```

Complexity Analysis

Algorithm	Best Case	Average Case	Worst Case
Linear Search	O(1)	O(n)	O(n)
Binary Search	O(1)	O(log n)	O(log n)

Conclusion

Linear Search is simple and works on any list.
Binary Search is faster for large datasets.
Binary Search requires the products to be sorted by Product ID.
For large e-commerce platforms, Binary Search is more efficient

Exercise 7: Financial Forecasting:

FinancialForecast.java

```
public class FinancialForecast {

  public static double forecastValue(
      double investment,
      double growthRate,
      int years) {

    if (years == 0) {
      return investment;
    }

    return forecastValue(
        investment * (1 + growthRate),
        growthRate,
        years - 1);
  }

  public static void main(String[] args) {

    double investment = 50000;
    double growthRate = 0.08;
    int years = 4;

    double futureValue =
        forecastValue(
            investment,
            growthRate,
            years);
```

```
        System.out.println(
            "Future Investment Value after "
            + years +
            " years = ₹" +
            futureValue);
    }
}
```

OUTPUT:

```
Future Investment Value after 4 years = ₹68024.44800000002
```